

RANJITH PACHAMUTHU

AI Software Engineer | Computer Vision | Generative AI | AI/ML

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CAREER OBJECTIVE

AI Engineer with 2+ years of professional experience developing computer vision, real-time AI, and intelligent software solutions. Seeking to leverage experience across AI/ML, computer vision, Generative AI, LLMs, and AI application development to build scalable, production-ready AI systems and contribute to innovative products solving real-world problems.

TECHNICAL SKILLS

Programming: Python, C++, Go

Computer Vision: OpenCV, YOLO, RetinaFace, MTCNN, AdaFace, FaceNet, FAISS, Object Detection, Face Recognition, Tracking, RTSP, ANPR

AI / ML: PyTorch, Deep Learning, Machine Learning, Transformers, Embeddings, Model Inference

OCR / Document AI: Tesseract, PaddleOCR, PyMuPDF, Image Processing, Document Parsing

Backend: FastAPI, Flask, REST APIs, Microservices, WebSockets, RBAC

Databases: MySQL, PostgreSQL, Redis, SQLite, SQL, ClickHouse, Vector Databases

Infrastructure: Docker, Linux, Kubernetes, Git, CUDA, ONNX Runtime, TensorRT

Generative AI: LLMs, RAG, AI Agents, LangChain, LangGraph, MCP, VLMs

AR/VR: Unity, Unreal Engine, Blender

PROFESSIONAL EXPERIENCE

Software / AI Developer — iProAT Solutions, Chennai

Sept 2024 – Present

- Handled end-to-end AI software development across backend engineering, system architecture, database design, computer-vision model development, deployment, and distributed infrastructure.
- Designed backend APIs, database schemas, and service architecture using Python, FastAPI/Flask, Laravel, MySQL/PostgreSQL, Redis, and Kafka.
- Developed and integrated computer-vision models for face recognition, PPE detection, and ANPR, covering model training, inference, validation, and optimization.
- Deployed and managed AI applications using Docker, Linux, CUDA, ONNX Runtime, TensorRT, and distributed/orchestration technologies.
- Architected scalable CCTV processing solutions supporting 2,000+ cameras through distributed node/cluster-based workload distribution.

Key Projects:

- AI CCTV Attendance & Face Recognition** — Built scalable real-time face detection, tracking, recognition, vector search, user enrollment, and visitor-management workflows.
- PPE Detection & Safety Monitoring** — Developed real-time CCTV-based PPE and safety-object detection for helmets, safety vests, and kitchen PPE, with restricted/unrestricted zone monitoring and real-time notification alerts for detected safety violations.
- ANPR & Parking Management** — Integrated ANPR cameras with vehicle identification, entry/exit tracking, visitor parking, and payment workflows.

Software Development Intern — Aroopa Technologies

Sep 2023 – May 2024

- Built an invoice OCR and document-parsing pipeline using Python, OpenCV, Tesseract/PaddleOCR, and PyMuPDF, transforming unstructured invoice documents into structured application data.
- Streamlined document processing through image preprocessing, OCR extraction, field parsing, validation, and handling of varying invoice layouts.

EDUCATION

B.Tech — Artificial Intelligence & Data Science

2020 – 2024

Sri Shanmugha College of Engineering and Technology